

GRANCEA ALEXANDRU-NICOLAE

SOFTWARE ENGINEER | iOS & DISTRIBUTED SYSTEMS

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PROFESSIONAL SUMMARY

Software Engineer at Garmin and Master's student in High Performance Computing & Big Data Analytics at Babeş-Bolyai University. Enterprise engineering experience across native mobile development (iOS/Swift) and cloud microservices. Algorithmic foundation backed by a National Informatics Olympiad Bronze Medal and programming instruction background. Adept at driving full-lifecycle feature engineering, performance optimization, and scalable distributed system integration.

WORK EXPERIENCE

Garmin — Software Engineer I (Xero Team)

Cluj-Napoca, Romania | Feb 2026 – Present

- Promoted from Mobile SWE Intern to full-time Software Engineer; architect and ship production-grade native features and reactive UI workflows for the **ShotView** iOS application.
- Drive end-to-end bug resolution, performance optimizations, and tier-3 customer escalations, ensuring robust stability and seamless device telemetry communication.
- Collaborate with cross-functional engineering teams in Agile sprints, leading feature planning, technical refinements, and high-standard peer code reviews.

Tech Stack: Swift, SwiftUI, UIKit, iOS SDK, Xcode, Bluetooth Telemetry, GRDB, Git, Agile/Scrum.

Amazon — Software Engineering Intern (Helis Team)

Iaşi, Romania | Jul 2025 – Oct 2025

- Engineered automated internal deployment policies utilizing **AWS Lambda (Java/TypeScript)**, **DynamoDB Streams**, and **SQS** to enhance pipeline reliability.
- Implemented high-throughput REST API endpoints to securely query and aggregate real-time service deployment logs from DynamoDB.
- Enhanced operational visibility by adding custom metrics to an internal React UI and configuring CloudWatch alarms for automated incident triage.

Tech Stack: Java, TypeScript, React, AWS Lambda, DynamoDB, DynamoDB Streams, SQS, CloudWatch.

Inventika Place — Programming Instructor (Part-Time)

Cluj-Napoca, Romania | Sep 2023 – Jul 2025

- Mentored students in advanced data structures, algorithmic complexity, dynamic programming, and graph theory in C/C++ for national competitive programming contests.

EDUCATION

Babeş-Bolyai University — M.Sc. in High Performance Computing & Big Data Analytics

Cluj-Napoca, Romania | Oct 2026 – 2028
(Expected)

- Focus:** Distributed computing, large-scale data processing architectures, parallel algorithms, and high-throughput systems.

Babeş-Bolyai University — B.Sc. in Computer Science (English Taught Program)

Cluj-Napoca, Romania | 2023 – 2026

- BSc Thesis:** Fine-tuning Whisper Model for Automatic Speech Recognition in Romanian (Speech ML pipeline & WER optimization).

KEY TECHNICAL PROJECTS

Event Ticket Platform — Java 21, Spring Boot, PostgreSQL, Flyway, JWT, Next.js

[GitHub](#)

- Engineered a layered REST API featuring JWT multi-role authorization (Organizers, Staff, Attendees), QR validation via ZXing, automated PDF generation via iText, and versioned database schema migrations with Flyway.
Gym Journal — React, Node.js, Express, PostgreSQL, Gemini API [Demo](#) / [GitHub](#)
- Built a full-stack workout intelligence platform with custom session analytics, interactive progress charting, and an integrated Gemini API assistant delivering tailored training insights.
Habit Tracker (CS50 Final Project) — Django, Python, SQLite, JavaScript [Demo](#) / [GitHub](#)
- Developed a routine-tracking application implementing secure session handling, dynamic streak visualization, and interactive habit completion metrics.

TECHNICAL SKILLS & CORE ACHIEVEMENTS

Languages:	Swift, Java, C/C++, Python, TypeScript, JavaScript, SQL, Kotlin, HTML/CSS.
Frameworks & Tools:	SwiftUI, UIKit, iOS SDK, Spring Boot, React, Next.js, Node.js/Express, Django, AWS (Lambda, DynamoDB, SQS, CloudWatch), Flyway, GRDB, Git.
Core Competencies:	Object-Oriented Design (OOP), Data Structures & Algorithms, RESTful APIs, Multithreading, CI/CD, Agile/Scrum.
Honors & Contests:	Bronze Medal – National Olympiad in Informatics; 3rd Place – National Algorithmic Contest; National Handball Olympiad Finalist; Harvard CS50 Certificate.
Languages Spoken:	Romanian (Native), English (B2).